Science of the Total Environment 789 (2021) 147939

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

Treatment of fresh leachate by microaeration pretreatment combined with IC-AO^2 process: Performance and mechanistic insight

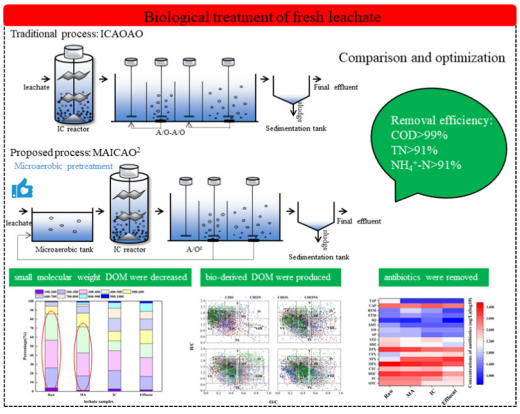
Mengqi Wei [a], Ruina Zhang [c,d], Mingjian Zhou [a], Zhiqiang Yuan [a], Haiping Yuan [a], Nanwen Zhu [a,b,*]

[a] School of Environmental Science & Engineering, Shanghai Jiao Tong University, Shanghai 200240, China
[b] Shanghai Engineering Research Center of Solid Waste Treatment and Resource Recovery, Shanghai 200240, China
[c] Shanghai Institute for Design & Research on Environmental Engineering Co., Ltd, Shanghai 200040, China
[d] Shanghai Environmental Sanitation Engineering Design Institute Co., Ltd, Shanghai 200001, China

HIGHLIGHTS

- MAICAO^2 process was innovatively proposed and showed better pollutant removal efficiency than traditional biological process.
- The removal efficiencies of COD, TN and NH_4^+-N in MAICAO^2 process increased 2%, 14% and 13% compared to ICAOAO process.
- ESI FT-ICR MS showed that microaeration ensured more than 53% of small molecular organic acids degraded in MA tank.
- The concentration variations of 18 antibiotics during MAICAO^2 process were studied.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 22 March 2021
Received in revised form 17 May 2021
Accepted 18 May 2021
Available online 23 May 2021

Editor: Yifeng Zhang

Keywords:
Fresh leachate
Biological treatment
Microaeration
Molecular information
Antibiotics

ABSTRACT

Fresh leachate is commonly featured with high concentrations of degradable organic matters, which can impede the performance of traditional biological treatment, especially the anaerobic reactor. Aiming at improving the biological treatment process of fresh leachate, this study creatively proposed a microaerobic-IC-AO^2 (MAICAO^2) process and compared it with traditional biological process, then optimized the operating conditions. Meanwhile, this work investigated the transformation rules and molecular compositions of dissolved organic matters (DOM) during MAICAO^2 process, particularly the hazardous DOM (antibiotics). The innovative MAICAO^2 process can effectively remove 99% chemical oxygen demand (COD), 91% total nitrogen (TN) and 91% ammonia (NH_4^+-N) during the operation time, and the removal efficiencies of COD, TN and NH_4^+-N in MAICAO^2 process increased approximately 2%, 14% and 13% compared to ICAOAO process. Electrospray ionization Fourier transform ion cyclotron resonance mass spectrometry (ESI FT-ICR MS) confirmed that microaeration could ensure over 53% small molecular organic acids degrade before the subsequent anaerobic reaction so the system could resist the high concentration organic matters stress and improve the denitrification efficiency. Further analysis showed that different categories of antibiotics (including 6 sulfonamides, 4 tetracyclines, 2 macrolides, 4 quinolones and 2 chloramphenicols) could be effectively removed by MAICAO^2 process with the total removal efficiency of 50%. This work proposed a new scenario for fresh leachate treatment by proposing the importance of the microaeration pretreatment during the biological treatment process.

© 2021 Elsevier B.V. All rights reserved.

* Corresponding author at: School of Environmental Science & Engineering, Shanghai Jiao Tong University, Shanghai 200240, China.
E-mail address: nwzhu@sjtu.edu.cn (N. Zhu).

https://doi.org/10.1016/j.scitotenv.2021.147939
0048-9697/© 2021 Elsevier B.V. All rights reserved.

1. Introduction

With the increasing speed of urbanization and the rapid growth of population, the production of municipal solid waste (MSW) is annually increasing (Hassan et al., 2016; Song et al., 2020). According to the data from the National Bureau of Statistic of China, the amount of MSW undergoing innocuous disposal reached 242 million metric tons (t) in 2020 (National Bureau of Statistics of China, 2021). As an effective method for the reduction of the MSW volume and harvest of the energy, incineration technology has developed rapidly in recent years (Li et al., 2017), and the proportion of incineration has reached 50% (National Bureau of Statistics of China, 2021). In order to ameliorate the corresponding heat value before MSW are incinerated to generate power, wastes are generally piled up and fermented in a pit for several days to take moisture away from the wastes (Chen et al., 2020). During this period, a great amount of fresh leachate could be generated, which accounts for approximately 15%–25% of all MSW (Dang et al., 2016; Zhou et al., 2017). The resulting leachate contains a complex composition with a high concentration of organic substances (Liu et al., 2015) as well as the refractory organics, for example humic and fulvic compounds (Dang et al., 2016). Considering the improper treatment of fresh leachate will cause irreversible damage to the environment (Zhang et al., 2019b), it is necessary to explore an effective method for fresh leachate treatment (Ye et al., 2011).

Currently, leachate are usually treated in biological and physicochemical methods (Yuan et al., 2017b). The physicochemical method can eliminate a large amount of organic matters in fresh leachate. Advanced oxidation is an effective method for wastewater treatment (Iovino et al., 2019; Salvestrini et al., 2020). Divyapriya et al. (2021) discussed the feasibility of photoelectrochemical used to disposal wastewater. Ozone and Fenton oxidation methods showed high micro-pollutants degradation potential (Ateş and Argun, 2021). However, they are expensive and prone to produce secondary pollution (Miao et al., 2019; Song et al., 2020). In contrast, the biological processes are extensively used to remove organic and nitrogen from leachate because of the cost-effectiveness and simplicity (Jokela et al., 2002). Due to the strict implementation of discharge criteria of leachate (GB 16889-2008) in China, a combined "anaerobic digestion (AD) + membrane bioreactor (MBR) + membrane-driven" process is widely used for leachate treatment (Wang et al., 2020). Nevertheless, in the traditional biological treatment process, the fast degradation of the organics such as lipids and proteins in leachate usually lead to the accumulation of volatile fatty acids (VFAs), therefore reducing the pH drastically (Carballa et al., 2011; Ding et al., 2021; Yuan et al., 2017a). The decrease of pH weakened the activity of microorganisms in the anaerobic reactor (Ding et al., 2021). Furthermore, the massive degradation of organic matters causes the shortage of carbon sources in the subsequent denitrification section (Fu et al., 2015). Although the effect of acidification could be offset by enlarging the reactor volume and reducing the influent load, the degradation of VFAs in AD greatly decreased the ratio of biochemical oxygen demand to total nitrogen (BOD/TN) of effluent, resulting in the decline of denitrification efficiency (Begum et al., 2018). The improvement of traditionally biological treatment process for fresh leachate is of great significance.

Recent studies have demonstrated that the removal efficiency of organic matters could be enhanced by recycling terminal mixture to microaerobic tank (Benson et al., 2007; Warith, 2002). He et al. (2015) studied the applying of microaeration, and found that the recirculation of terminal mixture to the microaerobic tank improved the stability of the leachate treatment system. The microaerobic tank could not only buffer the fluctuation of quality and quantity of influent, but also reduce the inhibitory impact of acidic pH and high concentrations of ammonium in recycled liquid on microorganisms.

Most of bio-refractory organics in fresh leachate predominantly exist as dissolved organic matter (DOM), which is a main reason for high chemical oxygen demand (COD) in effluent (Jiang et al., 2018; Li et al., 2017). The characteristics of DOM are tightly associated with the treatment and process improvement of leachate disposal. Therefore, the recognition of the transformation mechanisms of DOM in biological process is of great significance to improve leachate treatment process (Kunacheva and Stuckey, 2014; Wang and Chen, 2018). However, most of the analytical methods of the properties of DOM are ultraviolet-visible spectrometry, gas chromatograph-mass spectrometry and three-dimensional fluorescence, which cannot provide accurate information about transformation rules of DOM (Chen et al., 2019; Ruiz-Delgado et al., 2020; Xi et al., 2012). Compared with traditional analytical methods, high resolution mass spectrometry, for example electrospray ionization Fourier transform ion cyclotron resonance mass spectrometry (ESI FT-ICR MS), is shown to be a unique method to provide molecular information of wastewater (Kellerman et al., 2014; Yuan et al., 2017b; Zhang et al., 2019a). As the dominant sites for disposal of MSW, incineration plant might receive a variety of antibiotics from discarded pharmaceuticals, personal care products and medical wastes (Graham et al., 2011; Threedeach et al., 2012). Masoner et al. (2014) found that whether in raw or final leachate, antibiotics were generally detectable. Thus, how antibiotics, an important type of hazardous DOM, transformed in bio-process of fresh leachate is still unclear.

Therefore, this study innovatively proposed a microaerobic-IC-AO^2 process ($MAICAO^2$) in order to improve the pollutant removal efficiency in leachate treatment process, and unravel the decontamination mechanism. Notably, the primary objectives of this study were: (1) to study the removal efficiency of pollutant in $MAICAO^2$ process, and compared it with traditional biological process, then optimized the operating conditions including reflux ratio (R), reflux distribution ratio (RD), hydraulic retention time (HRT) and sludge retention time (SRT); (2) to identify the transformation rules and molecular compositions of DOM in leachate during $MAICAO^2$ process by ESI FT-ICR MS; and (3) to study the degradation characteristics of selected hazardous DOM, antibiotics.

2. Materials and methods

2.1. Start-up of $MAICAO^2$ system and sample collection

In order to build $MAICAO^2$ system and compare it with traditional biological treatment, two devices for leachate treatment were designed, as described in Fig. 1. One of which was a simulated traditional activated sludge system (ICAOAO, Fig. 1a), and the other was an improved system ($MAICAO^2$, Fig. 1b). The two devices were wrapped with silicone heat conductive pad to keep the temperature of the system at 35 °C. ICAOAO process consisted of two parts, including an IC reactor (12 L) and an anoxic-oxic-anoxic-oxic tank (8, 8, 8 and 12 L). In $MAICAO^2$ system, in order to take full use of the high concentration organic matters in the fresh leachate as the carbon source for denitrification, the first anoxic tank in the traditional process was set before the anaerobic reactor as a MA tank, and the terminal mixture was recycled to MA tank. In MA unit ($V = 8$ L), intermittent aeration was realized by relay to control dissolved oxygen (DO) < 0.5 mg/L. The working volume of IC reactor was 12 L. The AO^2 tank (28 L) consisted of three consecutive pools, including one denitrification pool ($V = 8$ L) and two nitrification pools ($V = 8$ and 12 L). The oxic pool was continuously aerated and the DO was controlled at 3–5 mg/L.

The fresh leachate used in this work was collected from the incineration plant that stored MSW (2000 t/d) for 5–7 days. The characteristics of raw leachate were shown in Table 1. In order to maintain the stability of the system, followed sludge acclimation method was used to cultivate the activated sludge for 60 days. The extra sludge produced from the system was collected for further disposal. Leachate samples were collected once every two days to evaluate the system performance, and 3 parallel samples were collected. The final value is the average of the 3 test results. All collected samples were filtered by 0.45 μm membrane at the first time, and kept at 4 °C in dark until used.

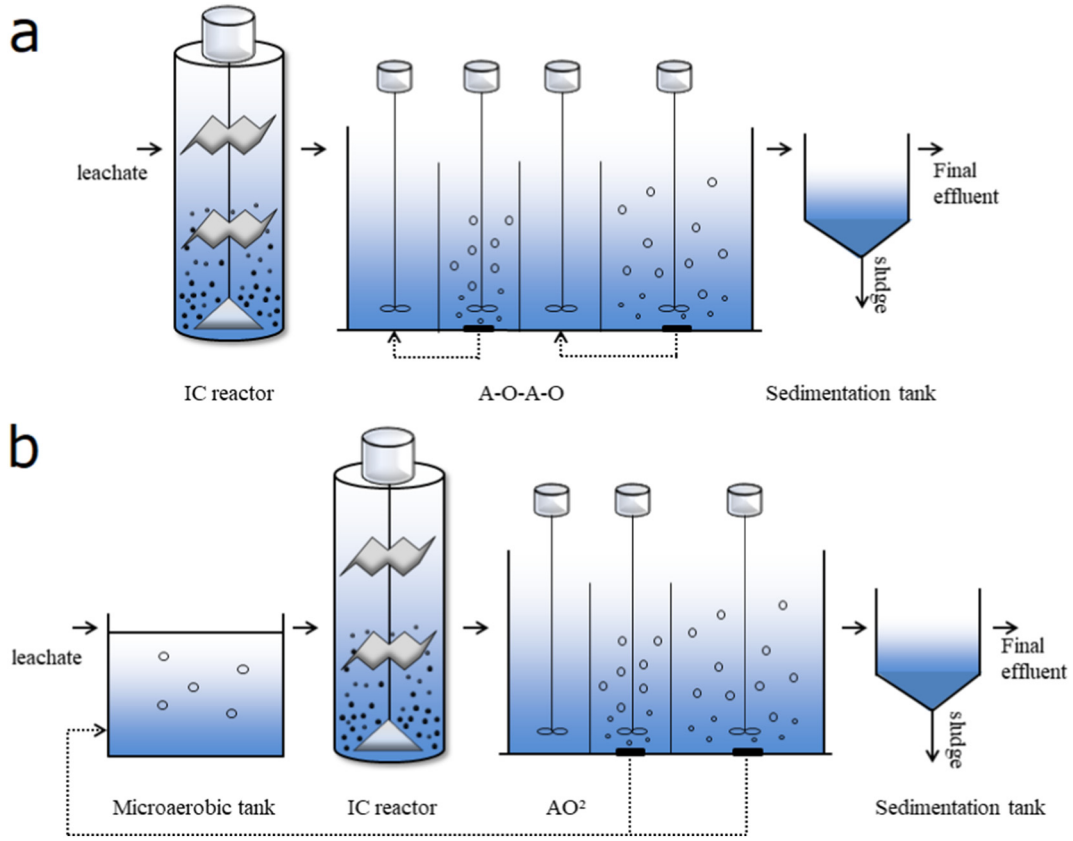


Fig. 1. Schematic diagram of the two designed reactors for leachate treatment. (a) Simulated traditional activated sludge system, ICAOAO, (b) proposed MAICAO^2 process.

2.2. Physio-chemical analysis

All reagents that were used in this study were analytical grade. DO was measured by portable dissolved oxygen meter (HQ40D, HACH Company, America). pH was measured by pH meter (PHS-25, Shanghai Leici Company, China). COD, BOD, TN, ammonia (NH_4^+-N), nitrate nitrogen (NO_3^--N) and suspended solid (SS) were measured in accordance with the Standard Methods (APHA, 1998). TOC was tested by TOC analyzer (TOC-V CPH/CPN, Shimadzu Co, Japan).

2.3. Molecular information acquisition by ESI FT-ICR MS and data analysis

In order to remove remaining salts and minimize matrix effects, solid phase extraction (SPE) process was used to pre-treatment samples. Leachate samples were collected once every day for 5 days at the optimal operating conditions, and the samples were mixed for DOM test. All filtered samples (200 mL) were acidified with hydrochloric acid to pH 2 and extracted by SPE. C_{18} cartridges (500 mg, 6 mL, Waters) were firstly activated with acid (pH = 2) and methanol (MS grade); Leachate samples were added to cartridges at a flow rate of 2 mL/min, then rinse the cartridges with 20 mL hydrochloric acid (pH = 2) to eliminate the salts; After that, cartridges were dried with a stream of N_2 and eluted with 1 mL of chromatographic grade methanol and kept at 4 °C in dark until tested.

A Bruker Apex Ultra ESI FT-ICR MS (Bruker Daltonics, Germany) was used to analyze samples at the condition of 7 T superconducting magnet. All samples were tested in negative ion mode, in which the ion accumulation, ESI voltages and number of co-added scans were optimized. A large amount of fragmented molecular ions were generated at atmospheric pressure coupled with negative ion mode (Fu et al., 2020; Sleighter et al., 2008). The criteria for eliminating molecular formula choices was shown below (Koch et al., 2007): (1) double bond equivalents (DBE) ≥ 0; (2) the sum of the number of hydrogen and nitrogen atoms should be even; (3) the number of carbon atoms must exceed (H − N − 2) / 2 and less than 3 times of the number of hydrogen atoms; and (4) the number of oxygen or nitrogen atoms should not exceed the number of carbon atoms.

DBE represents the number of rings and Π bonds in organic matter, which can be used to explain the degree of saturation of organic matter in mass spectrometry data. The calculation formula was shown below (Ohno et al., 2010):

$$DBE = 1 + 0.5 \times (2C - H + n) \quad (1)$$

According to van Krevelen diagram, leachate DOM could be divided into 9 categories by H/C and O/C cutoffs (Chen et al., 2018): (1) lipid-like (group I): O/C = 0–0.3, H/C = 1.5–2.5; (2) protein/peptides-like (group II): O/C = 0.3–0.55, H/C = 1.5–2.5; (3) amino sugars (group III): O/C = 0.55–0.67, H/C = 1.5–2.5; (4) carbohydrates (group IV): O/C = 0.67–1.2, H/C = 1.5–2.4; (5) other aliphatic (group V): including the area of I with no oxygen and the areas of II and III with no nitrogen; (6) unsaturated hydrocarbons (VI): O/C = 0–0.1, H/C = 0.7–1.5; (7) carboxylic rich alicyclic molecules (CRAM) or lignin-like (VII): O/C = 0.1–0.67, H/C = 0.7–1.5; (8) tannins (VII):

Table 1
The main characteristics of raw leachate.

Properties	Value
pH	6.2 ± 0.5
Suspended solid (SS, mg/L)	19,520 ± 4520
Chemical oxygen demand (COD, mg/L)	67,768 ± 8420
Total organic carbon (TOC, mg/L)	17,211 ± 4542
Biochemical oxygen demand (BOD_5, mg/L)	22,152 ± 5257
Total nitrogen (TN, mg/L)	2559 ± 426
Ammonia (NH_4^+-N, mg/L)	2353 ± 383

O/C = 0.67–1.2, H/C = 0.7–1.5; and (9) condensed aromatics (IX): O/C = 0–1.2, H/C = 0.2–0.7.

2.4. Quantification of antibiotics

Sample pretreatment and SPE method were used based on the previous studies (Hou et al., 2015; Luo et al., 2011), as described in the Text S1. Leachate samples were collected once every two days for 10 days at the optimal operating conditions and each sample was tested 3 times in the detection of antibiotics using Ultra High Performance Liquid Chromatography coupled with Triple Quad Mass (UHPLC-3Q-MS, Sciex, Japan). A total of 18 antibiotics, including 6 sulfonamides (SAs, sulfadiazine, sulfathiazole, sulfapyridine, sulfamerazine, sulfamethoxzole, sulfaquinoxaline), 4 tetracyclines (TCs, oxytetracycline, tetracycline, doxycycline, chlortetracycline), 2 macrolides (MLs, erythromycin, roxithromycin), 4 quinolones (FQs, ofloxacin, norfloxacin, ciprofloxacin, enrofloxacin), and 2 chloramphenicols (CPs, chloramphenicol, thiamphenicol) were detected. The physicochemical properties of the target antibiotics were presented in Table S1. Other information of analysis was detailed in the supplemental Text S2 and Table S2.

3. Results and discussion

3.1. General performance of MAICAO^2 process

3.1.1. Comparison of degradation performance on main pollutants between ICAOAO system and MAICAO^2 system

After 60 d-stabilization with the two designed systems at the same condition (R = 500%, RD = 1:1, HRT = 10 d, SRT = 15 d), the pollutant removal performance between ICAOAO process and MAICAO^2 process were compared, as shown in Fig. 2. The influent COD of the two processes fluctuated slightly, which were 66,281–71,116 mg/L, and the average COD of influent was 67,925 mg/L. The average COD of effluent was 2291 mg/L (ICAOAO process) and 1287 mg/L (MAICAO^2 process) respectively. Under the same operating conditions, the removal rate of COD in MAICAO^2 process was increased approximately 2% compared to ICAOAO process, indicating that MAICAO^2 system could decompose organic matters better. This is mainly because the MA tank could degrade some organic matters and a large amount of small molecular organic acids, then provided a suitable environment for subsequent reactions. The removal efficiency of COD in MA tank reached 83%. Compared with ICAOAO process, the MA tank reduced the impact load of the IC reactor, thus making the system more stable. Moreover, IC digesters have shown better removal efficiency of COD (>95%) compared with other digesters for example anaerobic membrane bioreactors (Arij et al., 2017; Lim and Kim, 2014; Svojitka et al., 2017).

After MAICAO^2 process, the average concentrations of TN, NH_4^+-N and NO_3^--N decreased from 2462, 2358 and 135 mg/L to 322, 242 and 17 mg/L, respectively. While after ICAOAO process, the average concentrations of TN, NH_4^+-N, and NO_3^--N decreased from 2426, 2295 and 129 mg/L to 521, 473 and 32 mg/L, respectively. The removal efficiencies of TN, NH_4^+-N and NO_3^--N in MAICAO^2 process were increased approximately 11%, 13% and 16% compared to ICAOAO process. The removal efficiency of TN reached 63% in MA tank. On one hand, the cycled terminal mixture diluted the organic concentration of the influent. On the other hand, simple organics such as small molecule proteins were absorbed by microorganisms in MA tank, so MA tank exhibited great potential for pollutant removal. Overall, it could be seen that MAICAO^2 process

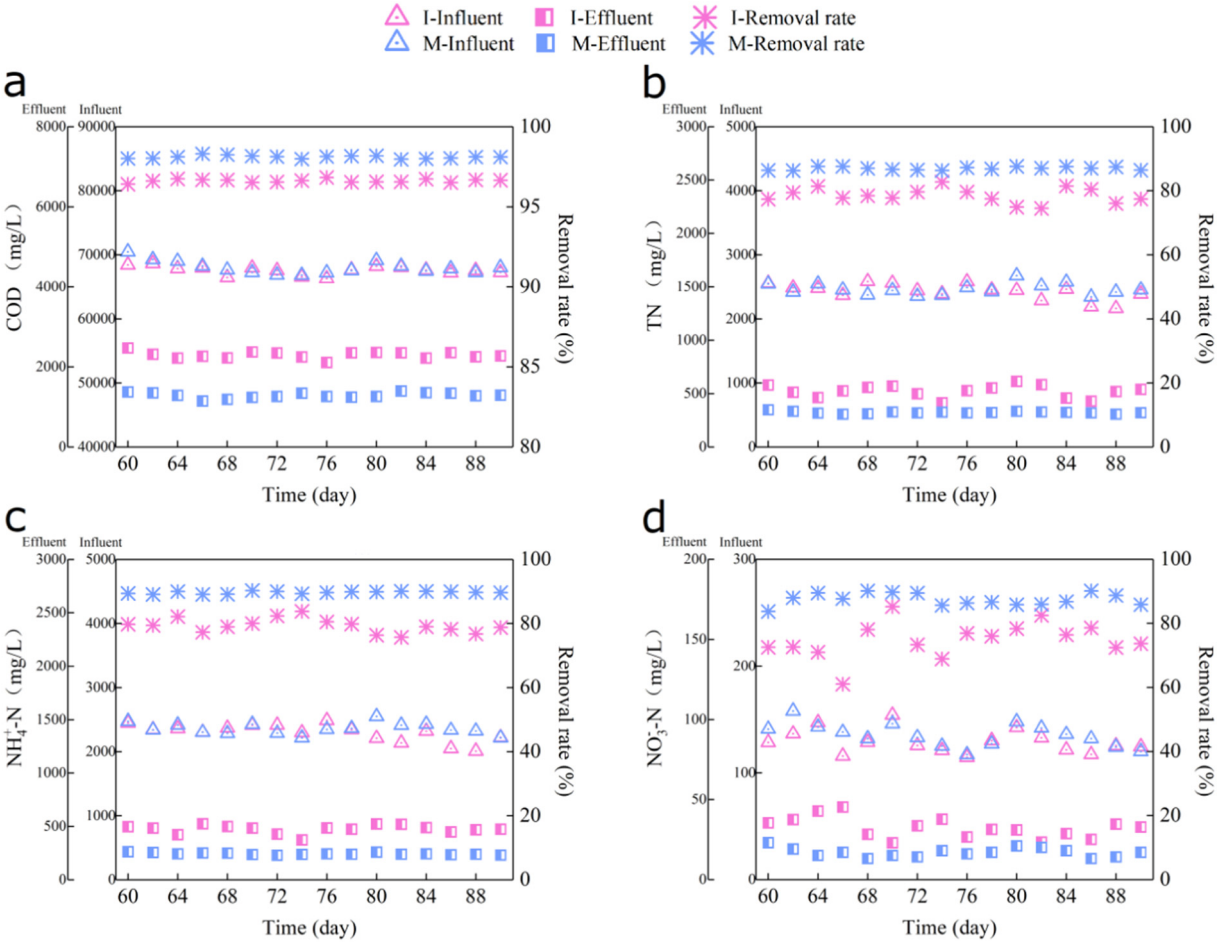


Fig. 2. Variations and removal efficiencies in terms of (a) COD, (b) TN, (c) NH_4^+-N, and (d) NO_3^--N by ICAOAO process (I) and MAICAO^2 process (M) under the same operating conditions.

was more effective in removing pollutants in the leachate compared to ICAOAO process. The recycling of the nitrification solution buffered the NH_4^+-N concentration of the influent, and part of the nitrogen was degraded in the MA tank, ensuring a suitable environment in the subsequent anaerobic reactor.

3.1.2. Optimization of operating conditions of MAICAO² process

The variations and removal efficiencies of COD, TN, NH_4^+-N and NO_3^--N from collected leachate samples of MAICAO² process were shown in Fig. 3. From day (d) 100 to day 330, the influence of R, RD, HRT and SRT were respectively identified. The specific operating conditions were shown in Table 2. The average concentrations of COD, TN, NH_4^+-N and NO_3^--N in the influent were 67,768, 2559, 2352 and 129 mg/L during the optimization process. Continuously optimization obtained the best operating conditions, and improved the removal performance of the main pollutants in the leachate by biological process.

In phase 1, 3 different gradients of R were controlled, successively 300%, 500%, and 700%. It could be seen from Fig. 3 that the removal efficiency of pollutants was the highest at R of 500%. The average concentrations of COD, TN, NH_4^+-N and NO_3^--N in the effluent were 1370, 320, 265 and 19 mg/L respectively, with the corresponding removal efficiencies of 98%, 87%, 89% and 85%. It's worth noting that at R of 700%, the foam in the oxic pool increased, and the activated sludge in the first oxic pool gradually turns black and the settling performance deteriorates. The increasing of R would provide more electron acceptors for the denitrification section, at the same time bring the DO into the denitrification system, thus destroying the anaerobic environment.

After determining the optimal R as 500%, the influence of the RD was explored. The distribution ratios of reflux from the first oxic pool and the second oxic pool to MA tank were set as 1:1, 3:2 and 4:1 respectively. As shown in Fig. 3, the removal efficiency was highest at RD of 3:2. The average concentrations of COD, TN, NH_4^+-N and NO_3^--N decreased from 67,025, 2426, 2352 and 124 mg/L to 1287, 264, 252 and 24 mg/L, and the removal efficiencies were 98%, 89%, 89% and 81% respectively. There was a significant difference in the removal efficiency under different RD, because the diverse RD would lead to different C/N ratio in denitrification section, resulting in great difference in nitrogen removal efficiency.

The HRT was a vital factor in the wastewater treatment process (Lusinier et al., 2020), and it was controlled by influent flow rate. Therefore, the effect of HRT (8, 10 and 12 d) on the process was explored during phase 3. Compared with the other two conditions, the pollutant removal efficiency was the highest at HRT of 12 d. The average

Table 2
The specific operating conditions during the period of optimization.

Period	R	RD	HRT (d)	SRT (d)
Phase 1	300% 500% 700%	1:1	10	15
Phase 2	500%	1:1 3:2 4:1	10	15
Phase 3	500%	3:2	8 10 12	15
Phase 4	500%	3:2	12	10 15 20

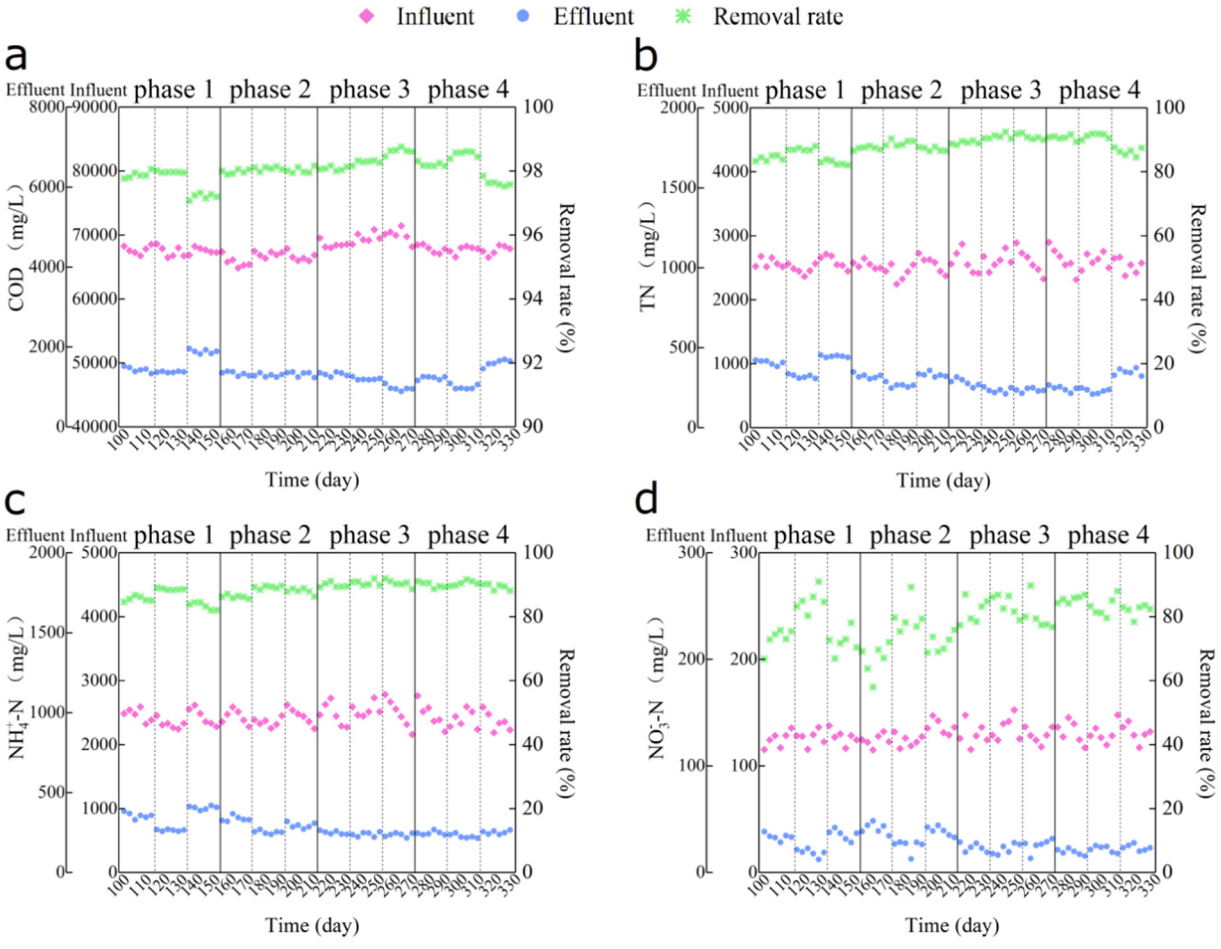


Fig. 3. Variations and removal efficiencies in terms of (a) COD, (b) TN, (c) NH_4^+-N, and (d) NO_3^--N by MAICAO² system. Leachate samples were collected once every two days after the operating parameters were adjusted for 10 days.

concentrations of COD, TN, NH_4^+-N and NO_3^--N in the effluent were 964, 234, 227 and 26 mg/L, with the removal efficiencies of 99%, 91%, 91% and 80% respectively. It seemed that the longer HRT, the higher removal efficiency of organic matters would be. However, excessive HRT would result in microorganisms being in the endogenous respiration period because of insufficient matrix content, which would decompose part of the intracellular or extracellular organic matter, and the removal rate would no longer increase but decrease (Fall et al., 2014).

In phase 4, the influence of SRT on the system was explored, successively 10, 15 and 20 d. The number of microorganisms was greatly decreased at SRT of 10 d, while the settling performance of activated sludge was observed to deteriorate at SRT of 20 d. Under normal condition, the concentration of SS in aerobic tank was maintained at 4000–6000 mg/L. However, the concentration of SS was tested as 5500–7000 mg/L at SRT of 20 d. The poor settling performance most likely resulted from high SS. The removal rate of pollutants reached the highest at SRT of 15 d. The average concentrations of COD, TN, NH_4^+-N and NO_3^--N in the effluent were 995, 229, 227 and 22 mg/L, with the corresponding removal rate of 99%, 91%, 91% and 83%. Song et al. (2020) studied the removal performance of pollutants by anoxic rotating biological contactor and four oxic tanks, and the removal efficiencies were 86%, 84% and 99% for COD, TN and NH_4^+-N. We conducted a multi-factor analysis of variance for the four parameters (R, RD, HRT and SRT), as shown in Table S3. The *p* values of the four parameters for COD, TN and NH_4^+-N were all less than 0.05, indicating that the four parameters have a significant impact on COD, TN and NH_4^+-N of the effluent. The p value of RD for NO_3^--N was less than 0.05 while the *p* values of R, HRT and SRT for NO_3^--N were higher than 0.05. This suggested that significant impact exist between RD and NO_3^--N, and R, HRT and SRT have no obvious effect on NO_3^--N.

From the above analysis, the optimal operating conditions were as follows: $R = 500\%$, RD = 2:3, HRT = 12 d and SRT = 15 d. After 330 d-treatment by the improved process, we reached relatively high removal efficiencies of COD > 99%, TN > 91%, NH_4^+-N > 91% in MAICAO^2 process, and the removal efficiencies were increased approximately 2%, 14% and 13% compared to ICAOAO process.

3.2. DOM transformation analysis by ESI FT-ICR MS

3.2.1. Molecular weight transformation of DOM

DOM is an important part of COD, whether COD can be further removed by improving the bio-process generally depends on the molecular weight and composition of DOM. Therefore, it is necessary to analyze the transformation mechanisms of DOM in biological treatment processes. Analyses and statistics of ESI FT-ICR MS data were used to recognize the molecular information of DOM in fresh leachate, including molecular weight distribution diagram, van Krevelen diagram and DBE diagram (Nguyen et al., 2012). The whole mass value was set at *m/z* 100–1000, while the sample mass spectra mainly focused on 100–500 *m/z* (Fig. S1). It could be demonstrated from the original mass spectra that the distribution of masses in MA effluent was more regular than that in raw leachate and the mass distribution towards higher. The normal distribution of DOM was obviously observed after the anaerobic reaction, illustrating that the types of organic matter increased. During the anaerobic process, microorganisms used the original organic matter as a carbon source to metabolize and produce bio-derived organic matter. ESI FT-ICR MS demonstrated the molecular weight distribution of leachate samples among fresh leachate, MA effluent, IC effluent and AO^2 effluent (Fig. 4). Percentage of *m/z* 100–400 decreased from 57% (raw leachate) to 27% (MA effluent), and about 53% small molecular organic acids degrade before the anaerobic reaction. As the process progressed, the distribution of DOM became increasingly normal, indicated that complex organics were gradually degraded into simple compounds. In the MA tank, the proportion of small molecule organics (especially small molecule organic acids) decreased, making small molecule organics truly participate in denitrification as carbon source. Meanwhile, small molecule organics were not accumulated in MA tank due to microaeration environment, which provided the suitable concentration of small molecule organics for continuous reaction.

3.2.2. Molecular composition transformation of DOM

van Krevelen diagram analysis is a crucial method to illustrate accurate molecular formulae. The diagram is element-ratio plots, and each dot in the diagram represents the molar ratio of oxygen to carbon (O/C) on the x-axis and molar ratio of hydrogen to carbon (H/C) on the y-axis for that specific formula. Based on O/C and H/C ratio, leachate DOM has been divided into 9 categories corresponding to 9 groups in van Krevelen diagram (Fig. 5). In order to acquire more information, DOM in leachate were categorized into 4 groups after further calculations (Nuerenberg et al., 2019), namely aliphatic compounds (CHO), phenolic compounds (CHON), polyphenols (CHOS) and polycyclic aromatics (CHONS). The major components in leachate were composed of these 4 groups of DOM with specific element ratio (Fu et al., 2020).

A total of 4098, 6049, 6481 and 7488 molecule formulas of DOM were detected after select among raw leachate, MA effluent, IC effluent and AO^2 effluent, respectively (Fig. 5a–d). It was worth noting that the numbers and types of DOM in the leachate increased along the process, which indicated that a great amount of bio-derived DOM were produced during biological process (Patriarca et al., 2018), and the number of components showed obvious difference. The majority of compounds in raw leachate were lipid-like (35%, group I), protein/peptides-like (22%, group II), and CRAM or lignin-like (35%, group VII). Considering raw leachate was initial product of hydrolytic acidification of MSW (Eggen et al., 2010; Kjeldsen et al., 2002), the transformation of multiple DOM was normal phenomenon. The proportion of lipid-like decreased from 35% (raw leachate) to 6% (AO^2 effluent), illustrating that microorganisms degraded large number of lipid-like compounds with high H/C and O/C ratio in biological treatment and Lipid-like compounds were highly biodegradable. In contrast, CRAM or lignin-like compounds increased from 35% (raw leachate) to 64% (AO^2 effluent), which suggested CRAM or lignin-like were bio-derived compounds. Compared with raw leachate, the composition of organic compounds in final effluent changed a lot, and the total content was higher and the relative intensity was strikingly increased. In MAICAO^2 system, some native DOM were degraded and a large amount of bio-derived DOM were produced.

Fig. 5 showed that DOM in the effluent of different units had obviously complex composition and composed of numerous organics after selected, and these compounds were concentrated in the 4 groups (CHO, CHON, CHOS, CHONS). After the treatment of biological process, bioavailable DOM (for example CHO) was substantially eliminated, and the relative abundance of other three classes increased dramatically. A large amount of CHON were produced in anaerobic process

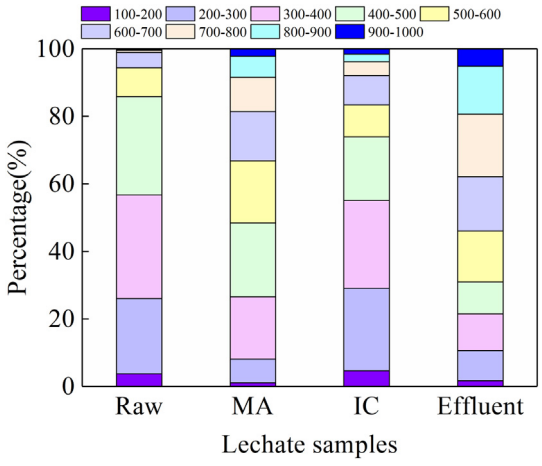


Fig. 4. Molecular weight distribution of leachate samples (Raw: raw leachate; MA: MA effluent; IC: IC effluent; Effluent: AO^2 effluent).

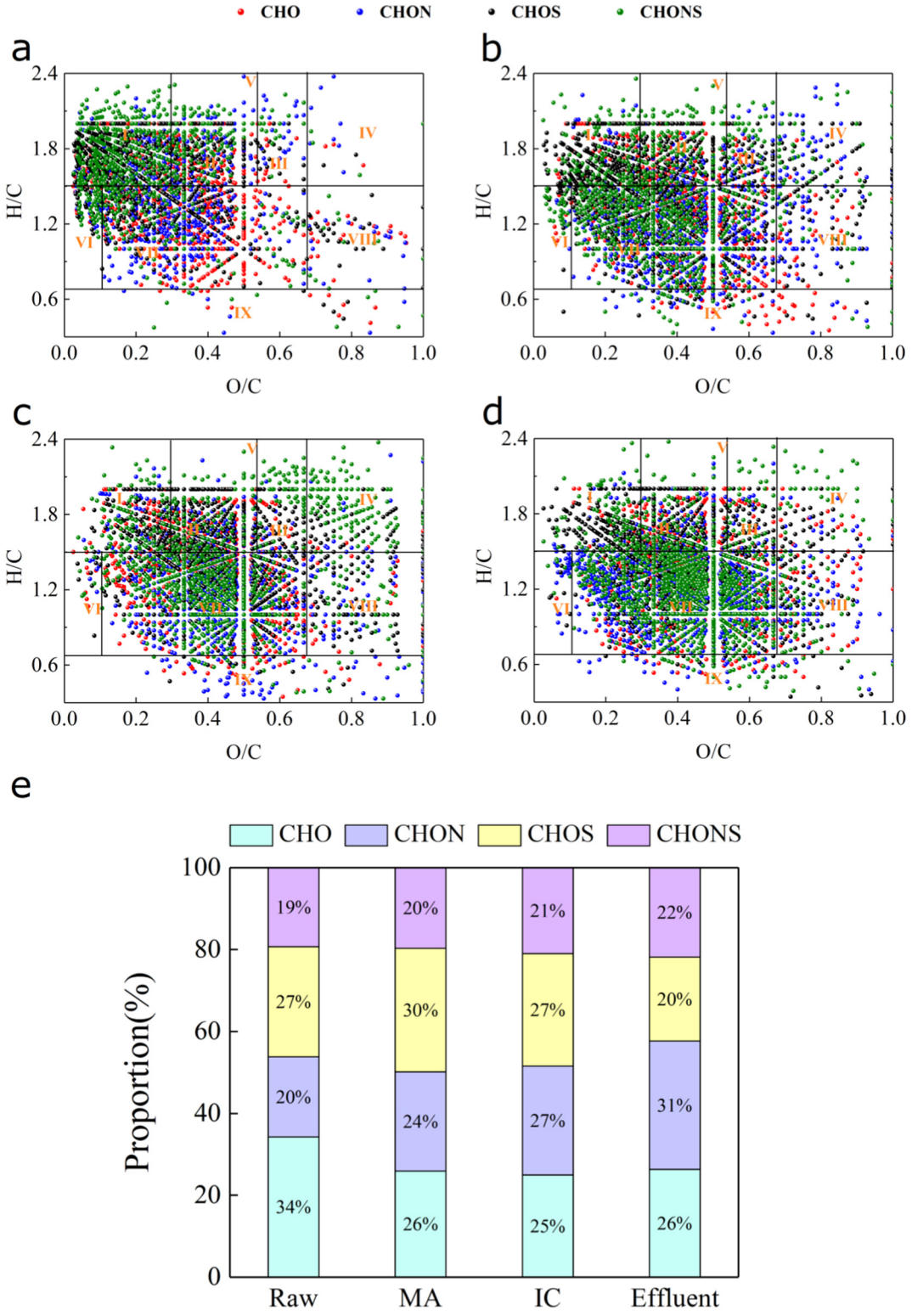


Fig. 5. Van Krevelen diagram of DOM in leachate. (a) Raw leachate, (b) MA effluent, (c) IC effluent, (d) AO^2 effluent, (e) subcategories distribution in leachate.

and they were mainly concentrated in group VII. Similarly, the AO^2 system was able to treat compounds with high bioavailability while producing DOM with more complex molecular compositions. The ESI TF-ICR MS results showed that there were obvious differences in the varieties of DOM substances between raw leachate and final effluent. The analysis of the van Krevelen diagram above demonstrated the biodegradation potential of these 4 DOM classes: CHO > CHOS > CHONS > CHON.

To better recognize the transformation mechanisms of DOM during MAICAO^2 process, double bond equivalent (DBE) was calculated, and it was vividly presented numbers of double bonds and rings in organic matters (Lv et al., 2016). The color of the dots in Fig. 6 corresponded to the O/C ratio. The majority of spots in Fig. 6a were red and orange, indicating that raw leachate contained a large number of compounds with lower degree of oxidation. There were more blue and green dots in final effluent, indicating that the degree of oxidation of organics had increased. As shown in

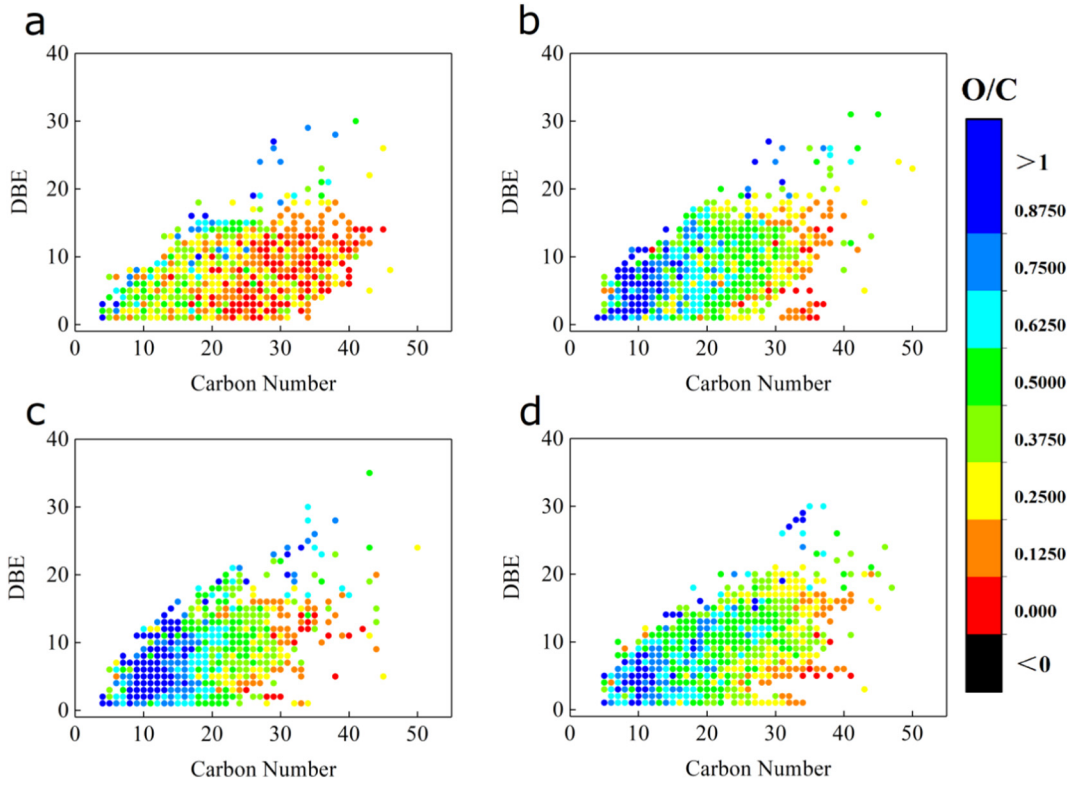


Fig. 6. Iso-abundance patterns of DBE versus number of carbons in leachate. (a) Raw leachate, (b) MA effluent, (c) IC effluent, (d) AO^2 effluent.

Fig. 6, the blue and green points are concentrated in the lower-left corner, which further shows that the DBE is positively correlated with unsaturation. DBE value could be influenced by the number of oxygen atoms as well as carboxyl groups consisted of oxygen atoms (Lv et al., 2016). As shown in Table S4, DOM in raw leachate has average DBE value of 6.1 and *m/z* value of 399.7, indicating a low unsaturation and high molecular weight. The above analysis showed that the raw leachate contains a lot of biodegradable compounds such as aliphatic compounds. By comparing the raw leachate and MA effluent, it could be found that the DBE value of the leachate increased significantly while the *m/z* decreased, indicating that a large amount of small molecular organic acids were produced. Slight increase of DBE_{avg} demonstrated the increase of unsaturated compounds and cyclic structures among $MAICAO^2$ processes. Compounds like ether and alcohol transformed to organics with carboxyl function groups under the oxidation of microorganisms.

3.3. Occurrence and distribution of selected hazardous DOM: antibiotics

Antibiotic, as a kind of hazardous DOM, has been regarded as environmental contaminants owing to associated risks exhibit to human, and it can get into the environment through incineration plant. Therefore, the concentration variation of antibiotics during biological treatment process was detected in this work. The average concentrations of all tested antibiotics and the concentrations of specific antibiotic in leachate sample were displayed in Fig. 7. A total of 18 antibiotics, including 6 sulfonamides (SQ, SMT, SM, SP, STZ and SDZ), 4 tetracyclines (CTC, DOC, TC and OTC), 2 macrolides (RTM and ETM), 4 quinolones (EFX, CFX, NFX and OFX), and 2 chloramphenicols (TAP and CAP) were detected in raw leachate. After $MAICAO^2$ process, the average concentration of 18 antibiotics decreased from 10.5 ± 0.8 to 5.1 ± 0.9 μg/L. All antibiotics could still be detected in the effluent, meaning that this

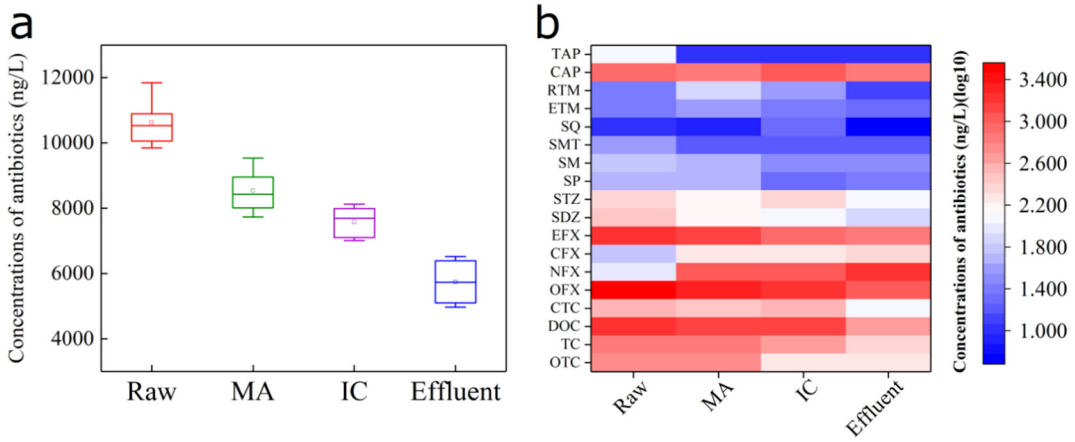


Fig. 7. Average concentrations of 18 antibiotics (a) and concentrations of specific antibiotics (b) in different units.

process could not completely remove antibiotics. The average removal rate of all antibiotics was about 50%, indicating the feasibility of this improved system for treating fresh leachate (Li et al., 2013).

Most detected antibiotics in raw leachate (TAP, CAP, RTM, ETM, SQ, SMT, SM, SP, STZ, SDZ, CFX, NFX, CTC, TC, OTC) were in the ng/L level, and relatively higher concentrations (μg/L level) were tested for EFX (1.8 ± 0.5 μg/L), OFX (3.6 ± 0.8 μg/L) and DOC (1.7 ± 0.6 μg/L). The dominant antibiotics in raw leachate were EFX, OFX and DOC, which account for about 67% of the total detected antibiotics. FQs and TCs were 2 dominant types of antibiotics in the influent with the average concentrations of 5.5 and 3.2 μg/L respectively, while the average concentrations of most subtypes of CPs, MLs and SAs were no more than 0.5 μg/L.

The analysis based on type suggested that the removal efficiencies of TCs and SAs were about 71% and 64%, and AO^2 reactors showed better removal performance compared to IC reactor (9% and 34%, respectively). The removal rate of TCs was the highest, and we speculate that the decrease of TCs was related to bio-sorption of activated sludge. The adsorption of TCs was usually a chemical process and dominated by ionic interaction, hydrophobicity, hydrogen bonding and cation bonding (Liao et al., 2021). The biodegradation of antibiotics was affected by the content of available DO, it is logical for IC reactor showed lower removal efficiency of antibiotics (Lin and Gan, 2011). Kang et al. (2018) studied the removal of SMX in anoxic-anaerobic-oxic system, and found that aerobic biodegradation was the major removal pathway while sorption and anoxic-anaerobic system were insignificant. Zhao et al. (2018) investigated the removal efficiency of nine SAs by anaerobic-anoxic-oxic-membrane system, and confirmed that aerobic biodegradation made the largest contribution to the removal of antibiotics. In our study, anaerobic reactor showed lower removal efficiency of 18 antibiotics compared to AO^2, and the conclusion were consistent with the results discussed above. Under aerobic condition, activated sludge was found capable of utilizing antibiotics as carbon source, and the adsorption performance of sludge was better compared to anaerobic condition. On the contrary, MLs, FQs and CPs were removed with lower efficiencies (41%, 42% and 38%, respectively). Overall, $MAICAO^2$ system exhibits great potential for the antibiotics removal, and the antibiotics were better removed by AO^2 reactor compared to IC reactor.

4. Conclusion

In this study, $MAICAO^2$ process was proven to be more efficient to simultaneously remove organic matters and nitrogen compared to ICAOAO process. $MAICAO^2$ system steadily performed for 330 d, and the average concentrations of COD, TN and NH_4^+-N decreased from 67,768, 2563 and 2433 mg/L to 995, 229 and 227 mg/L. The removal efficiencies of COD, TN and NH_4^+-N in $MAICAO^2$ process increased approximately 2%, 14% and 13% compared to ICAOAO process, respectively. Effluent recirculation, HRT and SRT showed to be important parameters to improve removal efficiencies, and the optimal operating conditions were: R = 500%, RD = 3:2, HRT = 12 d and SRT = 15 d.

High resolution mass spectrometry analysis indicated microaeration ensured more than 53% of small molecular organic acids degraded before the anaerobic reaction. About 3390 kinds of bio-derived DOM were produced during the process of biological treatment. Antibiotics, as a kind of hazardous DOM, could be effectively removed by $MAICAO^2$ process with the average removal rate of 50%. TCs and SAs were removed with high efficiencies (71% and 64%), while MLs, FQs, and CPs were worse removed (41%, 42% and 38% respectively). This study could provide a feasible idea for the improvement of fresh leachate treatment especially about the application of microaeration pretreatment.

CRediT authorship contribution statement

Mengqi Wei: Investigation, Conceptualization, Data curation, Writing – original draft, Writing – review & editing. **Ruina Zhang:** Formal analysis, Writing – review & editing, Resources. **Mingjian Zhou:** Formal analysis, Validation. **Zhiqiang Yuan:** Software, Writing – review & editing. **Haiping Yuan:** Project administration. **Nanwen Zhu:** Writing – review & editing, Funding acquisition, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was co-supported by the Foundation of Shanghai Municipal State-Owned Assets Supervision and Administration Commission (Grant No. 2018001) and the Shanghai Municipal Investment (Group) Corporation (CTKY-ZDXM-2020-015).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2021.147939.

References

APHA, 1998. Standard Methods for the Examination for Water and Wastewater, twenty first ed. American Public Health Association, Washington, D.C.
Arij, Y., Fatihah, S., Rakmi, A.R., Sarifah, Y., 2017. Optimization of operation conditions for the start-up of a pilot-scale anaerobic biofilm digester treating leachate. Desalin. Water Treat. 86, 43–50.
Ateş, H., Argun, M.E., 2021. Advanced oxidation of landfill leachate: removal of micropollutants and identification of by-products. J. Hazard. Mater. 413, 125326.
Begum, S., Anupoju, G.R., Sridhar, S., Bhargava, S.K., Jegatheesan, V., Eshtiaghi, N., 2018. Evaluation of single and two stage anaerobic digestion of landfill leachate: effect of pH and initial organic loading rate on volatile fatty acid (VFA) and biogas production. Bioresour. Technol. 251, 364–373.
Benson, C.H., Barlaz, M.A., Lane, D.T., Rawe, J.M., 2007. Practice review of five bioreactor/recirculation landfills. Waste Manag. 27 (1), 13–29.
Carballa, M., Duran, C., Hospido, A., 2011. Should we pretreat solid waste prior to anaerobic digestion? An assessment of its environmental cost. Environ. Sci. Technol. 45 (24), 10306–10314.
Chen, H., Yang, Z., Chu, R.K., Tolic, N., Liang, L., Graham, D.E., Wullschleger, S.D., Gu, B., 2018. Molecular insights into Arctic soil organic matter degradation under warming. Environ. Sci. Technol. 52 (8), 4555–4564.
Chen, W., Zhang, A., Jiang, G., Li, Q., 2019. Transformation and degradation mechanism of landfill leachates in a combined process of SAARB and ozonation. Waste Manag. 85, 283–294.
Chen, W., He, C., Zhuo, X., Wang, F., Li, Q., 2020. Comprehensive evaluation of dissolved organic matter molecular transformation in municipal solid waste incineration leachate. Chem. Eng. J. 400, 126003.
Dang, Y., Lei, Y., Liu, Z., Xue, Y., Sun, D., Wang, L.-Y., Holmes, D.E., 2016. Impact of fulvic acids on bio-methanogenic treatment of municipal solid waste incineration leachate. Water Res. 106, 71–78.
Ding, L., Chen, Y., Xu, Y., Hu, B., 2021. Improving treatment capacity and process stability via a two-stage anaerobic digestion of food waste combining solid-state acidogenesis and leachate methanogenesis/recirculation. J. Clean. Prod. 279, 123644.
Divyapriya, G., Singh, S., Martínez-Huitle, C.A., Scaria, J., Karim, A.V., Nidheesh, P.V., 2021. Treatment of real wastewater by photoelectrochemical methods: an overview. Chemosphere 276, 130188.
Eggen, T., Moeder, M., Arukwe, A., 2010. Municipal landfill leachates: a significant source for new and emerging pollutants. Sci. Total Environ. 408 (21), 5147–5157.
Fall, C., Rogel-Dorantes, J.A., Millan-Lagunas, E.L., Martinez-Garcia, C.G., Silva-Hernandez, B.C., Silva-Trejo, F.S., 2014. Modeling and parameter estimation of two-phase endogenous respirograms and COD measurements during aerobic digestion of biological sludge. Bioresour. Technol. 173, 291–300.
Fu, Q.-L., Fujii, M., Riedel, T., 2020. Development and comparison of formula assignment algorithms for ultrahigh-resolution mass spectra of natural organic matter. Anal. Chim. Acta 1125, 247–257.
Fu, S.-F., He, S., Shi, X.-S., Katukuri, N.R., Dai, M., Guo, R.-B., 2015. The chemical properties and microbial community characterization of the thermophilic microaerobic pretreatment process. Bioresour. Technol. 198, 497–502.
Graham, D.W., Olivares-Rieumont, S., Knapp, C.W., Lima, L., Werner, D., Bowen, E., 2011. Antibiotic resistance gene abundances associated with waste discharges to the Almendares River near Havana, Cuba. Environ. Sci. Technol. 45 (2), 418–424.
Hassan, M., Zhao, Y., Xie, B., 2016. Employing TiO_2 photocatalysis to deal with landfill leachate: current status and development. Chem. Eng. J. 285, 264–275.
He, R., Wei, X.-M., Tian, B.-H., Su, Y., Lu, Y.-L., 2015. Characterization of a joint recirculation of concentrated leachate and leachate to landfills with a microaerobic bioreactor for leachate treatment. Waste Manag. 46, 380–388.

Hou, J., Wan, W., Mao, D., Wang, C., Mu, Q., Qin, S., Luo, Y., 2015. Occurrence and distribution of sulfonamides, tetracyclines, quinolones, macrolides, and nitrofurans in livestock manure and amended soils of Northern China. Environ. Sci. Pollut. Res. Int. 22 (6), 4545–4554.
Iovino, P., Chianese, S., Prisciandaro, M., Musmarra, D., 2019. Triclosan photolysis: operating condition study and photo-oxidation pathway. Chem. Eng. J. 377, 121045.
Jiang, F., Qiu, B., Sun, D., 2018. Advanced degradation of refractory pollutants in incineration leachate by UV/peroxymonosulfate. Chem. Eng. J. 349, 338–346.
Jokela, J.P.Y., Kettunen, R.H., Sormunen, K.M., Rintala, J.A., 2002. Biological nitrogen removal from municipal landfill leachate: low-cost nitrification in biofilters and laboratory scale in-situ denitrification. Water Res. 36 (16), 4079–4087.
Kang, A.J., Brown, A.K., Wong, C.S., Yuan, Q., 2018. Removal of antibiotic sulfamethoxazole by anoxic/anaerobic/oxic granular and suspended activated sludge processes. Bioresour. Technol. 251, 151–157.
Kellerman, A.M., Dittmar, T., Kothawala, D.N., Tranvik, L.J., 2014. Chemodiversity of dissolved organic matter in lakes driven by climate and hydrology. Nat. Commun. 5, 3804.
Kjeldsen, P., Barlaz, M.A., Rooker, A.P., Baun, A., Ledin, A., Christensen, T.H., 2002. Present and long-term composition of MSW landfill leachate: a review. Environ. Sci. Technol. 32 (4), 297–336.
Koch, B.P., Dittmar, T., Witt, M., Kattner, G., 2007. Fundamentals of molecular formula assignment to ultrahigh resolution mass data of natural organic matter. Anal. Chem. 79 (4), 1758–1763.
Kunacheva, C., Stuckey, D.C., 2014. Analytical methods for soluble microbial products (SMP) and extracellular polymers (ECP) in wastewater treatment systems: a review. Water Res. 61, 1–18.
Li, C., Jiang, F., Sun, D., Qiu, B., 2017. Catalytic ozonation for advanced treatment of incineration leachate using (MnO_2-Co_3O_4)/AC as a catalyst. Chem. Eng. J. 325, 624–631.
Li, W., Shi, Y., Gao, L., Liu, J., Cai, Y., 2013. Occurrence and removal of antibiotics in a municipal wastewater reclamation plant in Beijing, China. Chemosphere 92 (4), 435–444.
Liao, Q., Rong, H., Zhao, M., Luo, H., Chu, Z., Wang, R., 2021. Interaction between tetracycline and microorganisms during wastewater treatment: a review. Sci. Total Environ. 757, 143981.
Lim, S.J., Kim, T.-H., 2014. Applicability and trends of anaerobic granular sludge treatment processes. Biomass Bioenergy 60, 189–202.
Lin, K., Gan, J., 2011. Sorption and degradation of wastewater-associated non-steroidal anti-inflammatory drugs and antibiotics in soils. Chemosphere 83 (3), 240–246.
Liu, Z., Dang, Y., Li, C., Sun, D., 2015. Inhibitory effect of high NH_4^+–N concentration on anaerobic biotreatment of fresh leachate from a municipal solid waste incineration plant. Waste Manag. 43, 188–195.
Luo, Y., Xu, L., Rysz, M., Wang, Y., Zhang, H., Alvarez, P.J.J., 2011. Occurrence and transport of tetracycline, sulfonamide, quinolone, and macrolide antibiotics in the Haihe River basin, China. Environ. Sci. Technol. 45 (5), 1827–1833.
Lusinier, N., Seyssiecq, I., Sambusiti, C., Jacob, M., Lesage, N., Roche, N., 2020. A comparative study of conventional activated sludge and fixed bed hybrid biological reactor for oilfield produced water treatment: influence of hydraulic retention time. Chem. Eng. J., 127611 https://doi.org/10.1016/j.cej.2020.127611.
Lv, J., Zhang, S., Wang, S., Luo, L., Cao, D., Christie, P., 2016. Molecular-scale investigation with ESI-FT-ICR-MS on fractionation of dissolved organic matter induced by adsorption on iron oxyhydroxides. Environ. Sci. Technol. 50 (5), 2328–2336.
Masoner, J.R., Kolpin, D.W., Furlong, E.T., Cozzarelli, I.M., Gray, J.L., Schwab, E.A., 2014. Contaminants of emerging concern in fresh leachate from landfills in the conterminous United States. Environ. Sci. Proc. Imp. 16, 2335–2354.
Miao, L., Yang, G., Tao, T., Peng, Y., 2019. Recent advances in nitrogen removal from landfill leachate using biological treatments — a review. J. Environ. Manag. 235, 178–185.
National Bureau of Statistics of China, 2021. National Bureau of Statistics of People's Republic of China, 2021. China Statistical Yearbook, 2021.
Nguyen, C., Elfland, C., Edwards, M., 2012. Impact of advanced water conservation features and new copper pipe on rapid chloramine decay and microbial regrowth. Water Res. 46 (3), 611–621.
Nuerenberg, G., Kunkel, U., Wick, A., Falas, P., Joss, A., Ternes, T.A., 2019. Nontarget analysis: a new tool for the evaluation of wastewater processes. Water Res. 163, 114842.
Ohno, T., He, Z., Sleighter, R.L., Honeycutt, C.W., Hatcher, P.G., 2010. Ultrahigh resolution mass spectrometry and indicator species analysis to identify marker components of soil- and plant biomass-derived organic matter fractions. Environ. Sci. Technol. 44 (22), 8594–8600.
Patriarca, C., Bergquist, J., Sjoberg, P.J.R., Tranvik, L., Hawkes, J.A., 2018. Online HPLC-ESI-HRMS method for the analysis and comparison of different dissolved organic matter samples. Environ. Sci. Technol. 52 (4), 2091–2099.
Ruiz-Delgado, A., Plaza-Bolaños, P., Oller, I., Malato, S., Agüera, A., 2020. Advanced evaluation of landfill leachate treatments by low and high-resolution mass spectrometry focusing on microcontaminant removal. J. Hazard. Mater. 384, 121372.
Salvestrini, S., Fenti, A., Chianese, S., Iovino, P., Musmarra, D., 2020. Electro-oxidation of humic acids using platinum electrodes: an experimental approach and kinetic modelling. Water 12 (8), 2250.
Sleighter, R.L., McKee, G.A., Liu, Z., Hatcher, P.G., 2008. Naturally present fatty acids as internal calibrants for Fourier transform mass spectra of dissolved organic matter. Limnol. Oceanogr. Meth. 6, 246–253.
Song, J., Zhang, W., Gao, J., Hu, X., Zhang, C., He, Q., Yang, F., Wang, H., Wang, X., Zhan, X., 2020. A pilot-scale study on the treatment of landfill leachate by a composite biological system under low dissolved oxygen conditions: performance and microbial community. Bioresour. Technol. 296, 122344.
Svojitka, J., Dvořák, L., Studer, M., Straub, J.O., Frömelt, H., Wintgens, T., 2017. Performance of an anaerobic membrane bioreactor for pharmaceutical wastewater treatment. Bioresour. Technol. 229, 180–189.
Threedeach, S., Chiemchaisri, W., Watanabe, T., Chiemchaisri, C., Honda, R., Yamamoto, K., 2012. Antibiotic resistance of Escherichia coli in leachates from municipal solid waste landfills: comparison between semi-aerobic and anaerobic operations. Bioresour. Technol. 113, 253–258.
Wang, H., Cheng, Z., Sun, Z., Zhu, N., Yuan, H., Lou, Z., Chen, X., 2020. Molecular insight into variations of dissolved organic matters in leachates along China's largest A/O-MBR-NF process to improve the removal efficiency. Chemosphere 243, 125354.
Wang, M., Chen, Y., 2018. Generation and characterization of DOM in wastewater treatment processes. Chemosphere 201, 96–109.
Warith, M., 2002. Bioreactor landfills: experimental and field results. Waste Manag. 22 (1), 7–17.
Xi, B.-D., He, X.-S., Wei, Z.-M., Jiang, Y.-H., Li, D., Pan, H.-W., Liu, H.-L., 2012. The composition and mercury complexation characteristics of dissolved organic matter in landfill leachates with different ages. Ecotoxicol. Environ. Safe. 86, 227–232.
Ye, J., Mu, Y., Cheng, X., Sun, D., 2011. Treatment of fresh leachate with high-strength organics and calcium from municipal solid waste incineration plant using UASB reactor. Bioresour. Technol. 102 (9), 5498–5503.
Yuan, C., Wang, Y., Zhu, T., Feng, X., Ma, Y., Yu, F., Xie, Y., 2017a. Multistage biological contact oxidation for landfill leachate treatment: optimization and bacteria community analysis. Int. Biodeterior. Biodegrad. 125, 200–207.
Yuan, Z., He, C., Shi, Q., Xu, C., Li, Z., Wang, C., Zhao, H., Ni, J., 2017b. Molecular insights into the transformation of dissolved organic matter in landfill leachate concentrate during biodegradation and coagulation processes using ESI FT-ICR MS. Environ. Sci. Technol. 51 (14), 8110–8118.
Zhang, B., Shan, C., Hao, Z., Liu, J., Wu, B., Pan, B., 2019a. Transformation of dissolved organic matter during full-scale treatment of integrated chemical wastewater: molecular composition correlated with spectral indexes and acute toxicity. Water Res. 157, 472–482.
Zhang, F., Peng, Y., Wang, S., Wang, Z., Jiang, H., 2019b. Efficient step-feed partial nitrification, simultaneous Anammox and denitrification (SPNAD) equipped with real-time control parameters treating raw mature landfill leachate. J. Hazard. Mater. 364, 163–172.
Zhao, W., Sui, Q., Mei, X., Cheng, X., 2018. Efficient elimination of sulfonamides by an anaerobic/anoxic/oxic-membrane bioreactor process: performance and influence of redox condition. Sci. Total Environ. 633, 668–676.
Zhou, L., Zhuang, W.-Q., Ye, B., De Costa, Y.G., Wang, X., Yu, K., Yi, S., Yang, S., Xia, S., 2017. Inorganic characteristics of cake layer in A/O MBR for anaerobically digested leachate from municipal solid waste incineration plant with MAP pretreatment. Chem. Eng. J. 327, 71–78.